"The Essentials of Computer Organization and Architecture"

Unit 6 – Exercises and Solutions

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ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

(*) complementary data:

- CPU Clock Frequency:
  500 MHz = 500 Mcycles/second = 500***10^6** cycles/second

- Polling Time:
  400 cycles/polling

- Mouse Polling Rate:
  30 pollings/second

- Floppy Disk Transfer Rate:
  50 Kbytes/s = 50***10^3** bytes/s, 16 bits at a time

- Hard Disk Transfer Rate:
  4 Mbytes/s = 4***10^6** bytes/s, 4 words (4*32bit) at a time

ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

i) CPU utilization when polling the mouse

Solution:

Assuming that mouse utilization is **100% (unrealistic)**

number of mouse pollings = Mouse Polling Rate = 30/sec
cycles per polling = Polling Time = 400 cycles/polling
number of cpu cycles = CPU Clock Frequency = 500Mcycles/sec

$$\textbf{CPU utilization} = \frac{\text{number of mouse pollings * cycles per polling}}{\text{number of cpu cycles}} * \mathbf{100\%}$$

= (30*400) / (500*10^6)
= 12000 / (500*10^6) = **0.0024%**

ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

ii) CPU utilization when polling the floppy disk drive

Solution:

Assuming that FDD utilization is **100% (unrealistic)**

Floppy Disk Transfer Rate = 50 Kbytes/s

num. of bytes transferred per polling = 2 bytes (16 bits)

$$\text{num. of floppy disk pollings} = \frac{\text{Floppy Disk Transfer Rate}}{\text{num. of bytes transferred per polling}}$$

$= (50*10^3)/2 = 25000$ pollings/sec

cycles per polling = Polling Time = 400 cycles/polling

number of cpu cycles = CPU Clock Frequency = 500Mcycles/sec

$$\textbf{CPU utilization} = \frac{\text{num. of floppy disk pollings.} * \text{cycles per polling}}{\text{number of cpu cycles}} * \mathbf{100\%}$$

$= (25000*400) / (500*10^6)$

$= (10*10^6) / (500*10^6) = \mathbf{2\%}$

ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

iii) CPU utilization when polling the hard drive

Solution:

Assuming that HDD utilization is **100% (unrealistic)**

Hard Disk Transfer Rate = 4 Mbytes/s = 4*10^6 bytes/s
num. of bytes transferred per polling = 16 bytes (4*32 bits)

$$\text{num. of hard disk pollings} = \frac{\text{Hard Disk Transfer Rate}}{\text{num. of bytes transferred per polling}}$$

=(4*10^6)/16 = 250000 pollings/sec

cycles per polling = Polling Time = 400 cycles/polling
number of cpu cycles = CPU Clock Frequency = 500Mcycles/sec

$$\textbf{CPU utilization} = \frac{\text{num. of hard disk pollings.} * \text{cycles per polling}}{\text{number of cpu cycles}} * \mathbf{100\%}$$

= (250000*400) / (500*10^6)
= (100*10^6) / (500*10^6) = **20%**

ES2 Find the CPU utilization when using interrupts to interact with a hard disk. [consider the information in (*)]

(*) complementary data:

- CPU Clock Frequency:
  500 MHz = 500 Mcycles/second = 500***10^6** cycles/second

- Interrupt Time:
  500 cycles/interrupt

- Disk Utilization: **100%** → for a fair comparison with polling ...

- Hard Disk Transfer Rate:
  4 Mbytes/s = 4***10^6** bytes/s, 4 words (4*32bit) at a time

ES2 Find the CPU utilization when using interrupts to interact with a hard disk. [consider the information in (*)]

CPU utilization when using interrupts to interact with the Disk

Solution:

In this scenario, using interrupts is not better than using polling, because handling one interrupt requires 100 more cycles than one polling … However, for Disk utilization bellow 100% (more realistic), using interrupts would largely pay off.

Hard Disk Transfer Rate = 4 Mbytes/s = 4*10^6 bytes/s

num. of bytes transferred per interrupt = 16 bytes (4*32 bits)

num. of hard disk interrupts = $\dfrac{\text{Hard Disk Transfer Rate}}{\text{num. of bytes transferred per interrupt}}$

=(4*10^6)/16 = 250000 interrupts/sec

cycles per interrupt = Interrupt Time = 500 cycles/interrupt

number of cpu cycles = CPU Clock Frequency = 500 Mcycles/sec

CPU utilization = $\dfrac{\text{num. of hard disk interrupts} * \text{cycles per interrupt}}{\text{number of cpu cycles}}$ * **100%**

= (250000*500) / (500*10^6)

= (125*10^6) / (500*10^6) = **25%**

ES3 Find the CPU utilization when using DMA transfers between hard disk and RAM. [consider the information in (*)]

(*) complementary data:

- CPU Clock Frequency:
  500 MHz = 500 Mcycles/second = 500***10^6** cycles/second
- "DMA Operation Start" Time:
  1000 cycles
- "DMA Operation End" Interrupt Time
  500 cycles
- DMA Operation Transfer:
  8 Kbytes = 8***10^3** bytes
- Hard Disk Transfer Rate:
  4 Mbytes/s = 4***10^6** bytes/s
- Disk Utilization: **100%** → for a fair comparison with polling and interrupts ...

ES3 Find the CPU utilization when using DMA transfers between hard disk and RAM. [consider the information in (*)]

CPU utilization when using DMA transfers

Solution:

In this scenario, the CPU utilization (**0.15%**) is MUCH lower than the one with polling (**20%**) or interrupts (**25%**).

Hard Disk Transfer Rate = 4 Mbytes/s = 4*10^6 bytes/s

num. bytes transfer. per DMA = DMA Operation Transfer = 8*10^3 bytes

$$\text{num. DMA transfers} = \frac{\text{Hard Disk Transfer Rate}}{\text{num. bytes transfer. per DMA}}$$

= (4*10^6)/8*10^3 = 500 transf./second

cycles per DMA = "DMA Operation Start" Time + "DMA Operation End" Interrupt Time = 1000+500 = 1500 cycles/DMA

number of cpu cycles = CPU Clock Frequency = 500Mcycles/sec

$$\textbf{CPU utilization} = \frac{\text{num. DMA transfers} * \text{cycles per DMA}}{\text{number of cpu cycles}} * 100\% =$$

= (500*1500) / (500*10^6)

= (750*10^3) / (500*10^6) = **0.15%**

ES4 A peripheral generates 4 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 500 MHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

Note: in this question, the symbols K, M, etc. refer to powers of 10.

a) The CPU interacts with the peripheral through polling. Each polling operation consumes 400 CPU clock cycles and is capable of collecting 16 bytes of data from the peripheral.

a1) The number of pollings per second required to sustain the peripheral throughput is: ____ (integer value)

a2) The number of CPU cycles spent to make the pollings calculated in a1) is: ____ (integer value)

a3) The CPU occupation rate (in %) represented by the value calculated in a2) is : ____ % (value to the hundredths, truncated)

ES4 A peripheral generates 4 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 500 MHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

a) The CPU interacts with the peripheral through polling. Each polling operation consumes 400 CPU clock cycles and is capable of collecting 16 bytes of data from the peripheral.

Problema Data:
A = peripheral transfer rate = 4 Mbytes / s = 4x10^6 bytes/s
B = CPU clock frequency = 500 MHz
= number of CPU clock cycles per second = 500x10^6 cycles/s
C = number of CPU cycles consumed per polling = 400 cycles
D = number of bytes transferred per polling = 16 bytes

Resolution:
E = number of pollings necessary to transfer 4 Mbytes in a second
E = A / D = 4 M / 16 = 4x10^6 / 16 = 10^6 / 4 = 0.25 x 10^6 = **250 000** pollings
F = number of CPU cycles spent in E pollings = E x C = 250 000 x 400 = **10^8** cycles
G = CPU occupation rate = F / B = 10^8 / (500x10^6) = 10^2 / 500 = 1/5 = **0.2**

Answer to a1): **250000** (integer value)
Answer to a2): **100000000** (integer value)
Answer to a3): **20.00** **%** (value to the hundredths, truncated)

ES4 A peripheral generates 4 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 500 MHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

Note: in this question, the symbols K, M, etc. refer to powers of 10.

b) The CPU interacts with the peripheral through interrupts generated by the peripheral. Each interrupt handling consumes 500 CPU clock cycles and is capable of collecting 32 bytes of data from the peripheral.

b1) The number of interrupts per second required to sustain the peripheral throughput is: ____ (integer value)

b2) The number of CPU cycles spent to handle the interrupts calculated in b1) is: ____ (integer value)

b3) The CPU occupation rate (in %) represented by the value calculated in b2) is : ____ % (value to the hundredths, truncated)

ES4 A peripheral generates 4 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 500 MHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

b) The CPU interacts with the peripheral through interrupts generated by the peripheral. Each interrupt handling consumes 500 CPU clock cycles and is capable of collecting 32 bytes of data from the peripheral.

Problema Data:

A = peripheral transfer rate = 4 Mbytes / s = 4x10^6 bytes/s
B = CPU clock frequency = 500 MHz
 = number of CPU clock cycles per second = 500x10^6 cycles/s
C = number of CPU cycles consumed per interrupt = 500 cycles
D = number of bytes transferred per interrupt = 32 bytes

Resolution:

E = number of interrupts necessary to transfer 4 Mbytes in a second
E = A / D = 4 M / 32 = 4x10^6 / 32 = 10^6 / 8 = 0.125 x 10^6 = **125 000** interrupts
F = number of CPU cycles spent in E interrupts = E x C = 125 000 x 500 = **62.5x10^6** cycles
G = CPU occupation rate = F / B = (62.5x10^6) / (500x10^6) = 62.5 / 500 = **0.125**

Answer to b1): **125000** (integer value)
Answer to b2): **62500000** (integer value)
Answer to b3): **12.50 %** (value to the hundredths, truncated)

ES4 A peripheral generates 4 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 500 MHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

Note: in this question, the symbols K, M, etc. refer to powers of 10.

c) The CPU uses a DMA controller to manage the interaction with the peripheral. The start and end of a DMA operation consumes 1000 and 500 CPU clock cycles respectively. Each DMA operation collects 8 KBytes of data from the peripheral.

c1) The number of DMA operations per second required to sustain the peripheral throughput is: ____ (integer value)

c2) The number of CPU cycles spent to start and end DMA operations calculated in c1) is: ____ (integer value)

c3) The CPU occupation rate (in %) represented by the value calculated in c2) is : ____ % (value to the hundredths, truncated)

ES4 A peripheral generates 4 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 500 MHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

c) The CPU uses a DMA controller to manage the interaction with the peripheral. The start and end of a DMA operation consumes 1000 and 500 CPU clock cycles respectively. Each DMA operation collects 8 KBytes of data from the peripheral.

Problema Data:
A = peripheral transfer rate = 4 Mbytes / s = 4x10^6 bytes/s
B = CPU clock frequency = 500 MHz
= number of CPU clock cycles per second = 500x10^6 cycles/s
Cs = number of CPU cycles consumed to start a DMA operation = 1000 cycles
Ce = number of CPU cycles consumed to end a DMA operation = 500 cycles
D = number of bytes transferred per DMA operation = 8 Kbytes = 8x10^3 bytes

Resolution:
E = number of DMA operations necessary to transfer 4 Mbytes in a second
E = A / D = 4 M / 8K = 0.5 x K = 0.5 x 1000 = **500** DMA operations
F = number of CPU cycles spent in E DMA operations = E x (Cs+Ce) = 500 x 1500 = **0.75x10^6** cycles
G = CPU occupation rate = F / B = (0.75x10^6) / (500x10^6) = 0.75 / 500 = **0.0015**

Answer to c1): **500** (integer value)
Answer to c2): **750000** (integer value)
Answer to c3): **0.15 %** (value to the hundredths, truncated)

HD1 A non-ZBR hard disk has 4 surfaces, 1024 tracks per surface, 128 sectors per track, 512 bytes per sector, a seek time of 5 milliseconds, and a rotational speed of 5000 RPM.

a) What is the capacity of the drive (in MBytes) ?

b) What is the access time (in ms) ?

HD1 A non-ZBR hard disk has 4 surfaces, 1024 tracks per surface, 128 sectors per track, 512 bytes per sector, a seek time of 5 milliseconds, and a rotational speed of 5000 RPM.

a) What is the capacity of the drive (in MBytes) ?

Solution:

4 surfaces x
1024 traces/surface x
128 sectors/track x
512 bytes/sector = 4 x 1024 x 128 x 512 = 268435456 bytes
= 2^2 x 2^10 x 2^7 x 2^9 = **2^28 bytes**
= 2^28 / **10^6** MBytes = **268,43 MBytes**

Hard drive capacity is measured in powers of 10 !

HD1 A non-ZBR hard disk has 4 surfaces, 1024 tracks per surface, 128 sectors per track, 512 bytes per sector, a seek time of 5 milliseconds, and a rotational speed of 5000 RPM.

b) What is the access time (in ms) ?

Solution:

rotational latency:

5000 rot ------ 60 s

1 rot ------ x s

x = 60 / 5000 = 0.012 s

$\bar{x}$ = x / 2 = 0.006 s = **6 ms**

seek time = 5 ms

rotational latency = [(60 s / 5000 rpm) x (1000 ms / s)] / 2 = **6 ms**

access time = 5 ms + 6 ms = 11 ms

HD2 A non-ZBR hard disk has 10 cylinders, 4 heads, 100 sectors per track, and 512 bytes per sector.

a) What is the capacity of the drive (in MBytes) ?

b) The LBA index of a sector is 1234. Compute the CHS (Cylinder,Head,Sector) coordinates of that sector.

LBA = (C × HPC + H) × SPT + (S − 1)
C = LBA div (HPC × SPT)
H = (LBA div SPT) mod HPC
S = (LBA mod SPT) + 1
HPC = Heads Per Cylinder (Surfaces)
SPT = Sectors Per Track

HD2 A non-ZBR hard disk has 10 cylinders, 4 heads, 100 sectors per track, and 512 bytes per sector.

a) What is the capacity of the drive (in MBytes) ?

Solution:

4 heads => 4 surfaces
10 cylinders => 10 tracks per surface

capacity of the drive = 4 surfaces x 10 tracks per surface x 100 sectors per track x 512 bytes per sector
= 4000 x 512 = 2048000 bytes
= 2048000 / 10^6 MBytes = **2,048 MBytes**

HD2 A non-ZBR hard disk has 10 cylinders, 4 heads, 100 sectors per track, and 512 bytes per sector.

b) The LBA index of a sector is 1234. Compute the CHS (Cylinder,Head,Sector) coordinates of that sector.

Solution:

LBA = (C × HPC + H) × SPT + (S − 1)
C = LBA div (HPC × SPT)
H = (LBA div SPT) mod HPC
S = (LBA mod SPT) + 1
HPC = Heads Per Cylinder (Surfaces)
SPT = Sectors Per Track

HPC = 4; SPT = 100
C = 1234 div (4 x 100) = 1234 div 400 = 3
H = (1234 div 100) mod 4 = 12 mod 4 = 0
S = (1234 mod 100) + 1 = 34 + 1 = 35

confirmation: LBA = (3 x 4 + 0) x 100 + (35 – 1) = 12 x 100 + 34 = 1234

HD3 A non-ZBR hard disk has 20 cylinders, 6 heads, 1800 sectors per track, and 4096 bytes per sector.

a) What is the capacity of the drive (in GBytes) ?

b) The LBA index of a sector is 123456. Compute the CHS (Cylinder,Head,Sector) coordinates of that sector.

LBA = (C × HPC + H) × SPT + (S − 1)
C = LBA div (HPC × SPT)
H = (LBA div SPT) mod HPC
S = (LBA mod SPT) + 1
HPC = Heads Per Cylinder (Surfaces)
SPT = Sectors Per Track

HD3 A non-ZBR hard disk has 20 cylinders, 6 heads, 1800 sectors per track, and 4096 bytes per sector.

a) What is the capacity of the drive (in GBytes) ?

Solution:

6 heads => 6 surfaces
20 cylinders => 20 tracks per surface

disk capacity = 6 surfaces x 20 tracks per surface x
1800 sectors per track x 4096 bytes/sector
= 884736000 bytes
= 884736000/10^9 GBytes = **0,884736 GBytes**

HD3 A non-ZBR hard disk has 20 cylinders, 6 heads, 1800 sectors per track, and 4096 bytes per sector.

b) The LBA index of a sector is 123456. Compute the CHS (Cylinder,Head,Sector) coordinates of that sector.

Solution:

LBA = (C × HPC + H) × SPT + (S − 1)
C = LBA div (HPC × SPT)
H = (LBA div SPT) mod HPC
S = (LBA mod SPT) + 1
HPC = Heads Per Cylinder (Surfaces)
SPT = Sectors Per Track

HPC = 6; SPT = 1800
C = 123456 div (6 x 1800) = 123456 div 10800 = 11
H = (123456 div 1800) mod 6 = 68 mod 6 = 2
S = (123456 mod 1800) + 1 = 1056 + 1 = 1057

confirmation: LBA = (11 x 6 + 2) x 1800 + (1057 – 1) = 68 x 1800 + 1056 = 123456

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let X=3, Y=100, Z=1000, W=512, R=4900 e T=0.005.

a) Indicate the total disk capacity, expressed in MBytes.

b1) Indicate the time (in milliseconds) it takes, on average, to start reading the first desired sector of a certain track, assuming that the read/write head is already positioned over that track.

b2) Solve b1) again, now assuming that the read/write head is not yet positioned over the correct track.

c) Indicate the number of read/write heads and cylinders on this hard drive.

d) Indicate the CHS coordinates equivalent to the coordinate LBA=12345.

LBA = (C × HPC + H) × SPT + (S − 1) ; **C** = LBA div (HPC × SPT)
H = (LBA div SPT) mod HPC ; **S** = (LBA mod SPT) + 1
HPC = Heads Per Cylinder (Surfaces) ; **SPT** = Sectors Per Track

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let X=3, Y=100, Z=1000, W=512, R=4900 e T=0.005.

a) Indicate the total disk capacity, expressed in MBytes.

Resolution:

disk capacity = X platters * 2 surfaces per platter * Y tracks per surface * Z sectors per track * W bytes per sector = 3 * 2 * 100 * 1000 * 512 = 307200000 bytes = 307200000 / 10^6 Mbytes = **307.2** MBytes

Answer: **307.20** MBytes (value to hundredths, truncated)

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let X=3, Y=100, Z=1000, W=512, R=4900 e T=0.005.

b1) Indicate the time (in milliseconds) it takes, on average, to start reading the first desired sector of a certain track (assuming that the read/write head is already positioned over that track).

Resolution:

The time requested is nothing else than the (average) rotational delay, which us half of the time needed for a disk rotation (turn).

Time of one rotation = 60s / R rotations = 60 / 4900 = 0.012244... s.

Time of half a rotation = 0.012244... / 2 = 0.006122... s = 6.122… ms

Answer: 6 milliseconds (integer part)

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let X=3, Y=100, Z=1000, W=512, R=4900 e T=0.005.

b2) Solve b1) again, now assuming that the read/write head is not yet positioned over the correct track.

Resolution:

In addition to the rotational latency (see b1)), it is also necessary to include the positioning time (seek time). In this way, the (average) access time becomes
T (seek time) + 0.006 (rotational latency) = 0.005 + 0.006122 = 0.011122 ... s

Answer: 11 milliseconds (integer part)

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let X=3, Y=100, Z=1000, W=512, R=4900 e T=0.005.

c) Indicate the number of read/write heads and cylinders on this hard drive.

Resolution:

One head is required for each surface. If there are X platters, "with all surfaces used", then there are X * 2 = 3 * 2 = 6 surfaces, and so there are **6** heads.

There are as many cylinders as there are tracks per surface. Therefore, with Y=100 tracks per surface, there are **100** cylinders.

Answer: **6** heads ; **100** cylinders

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let X=3, Y=100, Z=1000, W=512, R=4900 e T=0.005.

d) Indicate the CHS coordinates equivalent to the coordinate LBA=12345.

LBA = (C × HPC + H) × SPT + (S − 1) ; **C** = LBA div (HPC × SPT)
H = (LBA div SPT) mod HPC ; **S** = (LBA mod SPT) + 1
HPC = Heads Per Cylinder (Surfaces) ; **SPT** = Sectors Per Track

Resolution:

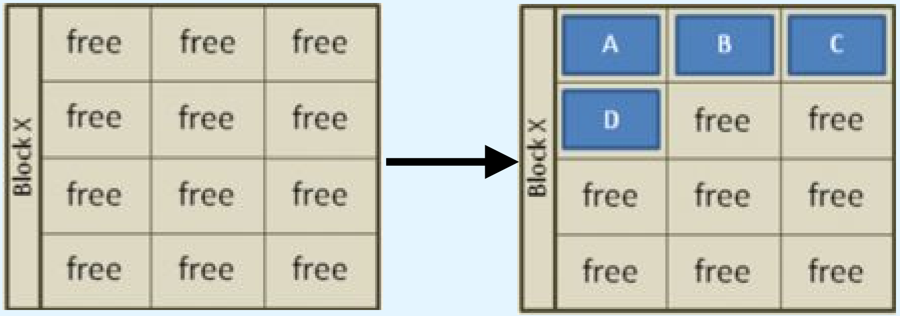
HPC = number of surfaces = X * 2 = 3 * 2 = 6 (see c)) ; SPT = Z = 1000
C = 12345 div (6 * 1000) = 12345 div 6000 = 2
H = (12345 div 1000) mod 6 = 12 mod 6 = 0
S = (12345 mod 1000) + 1 = 345 + 1 = 346

Confirmation:
LBA = (2 * 6 + 0) * 1000 + (346 - 1) = 12 * 1000 + 345 = 12000 + 345 = 12345

Answer: C = **2** ; H = **0** ; S = **346**

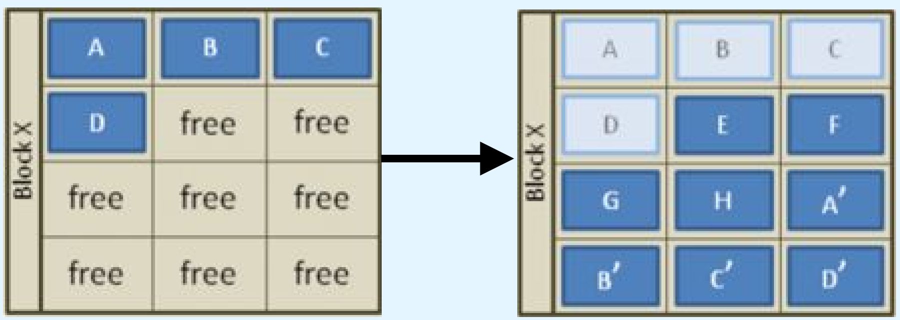
SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:

a)



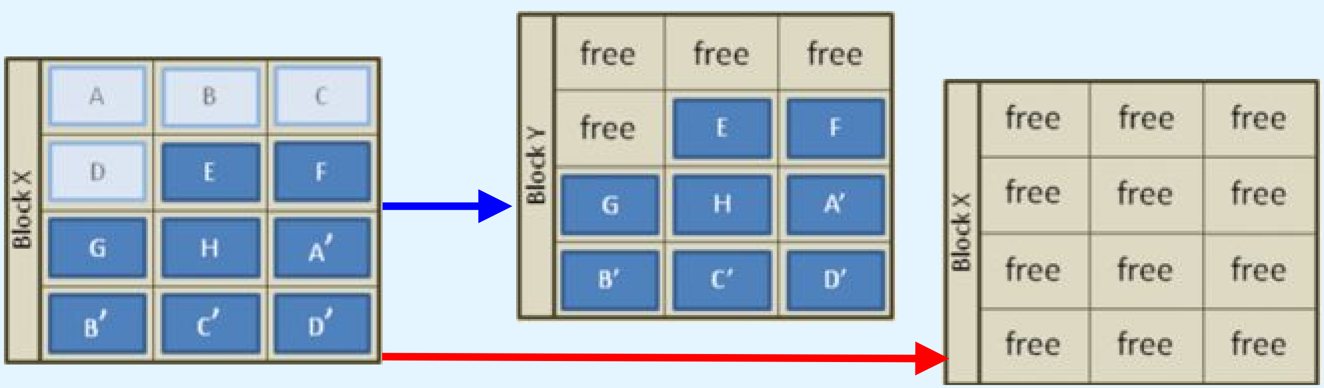
1. Four pages (A-D) are written to a block (X). Individual pages can be written at any time if they are currently free (erased).

b)



2. Four new pages (E-H) and four replacement pages (A′-D′) are written to the block (X). The original A-D pages are now invalid (stale) data, but cannot be overwritten until the whole block is erased.

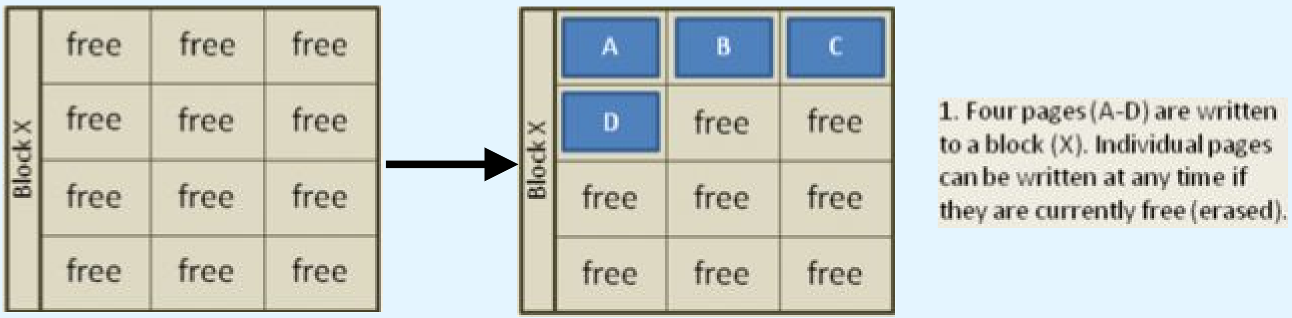
c)



3. In order to write to the pages with stale data (A-D) all good pages (E-H & A′-D′) are read and written to a new block (Y) then the old block (X) is erased. This last step is *garbage collection*.

SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:

a)

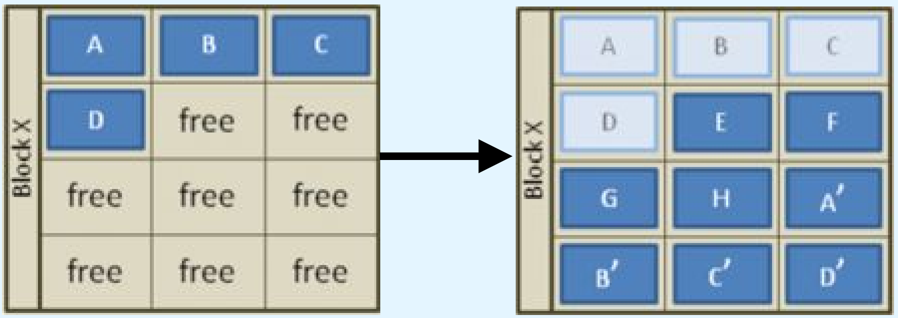


Solution:

T = 4 x 250 us (write A,B,C,D) = 1000 us = 1 ms

SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:

b)



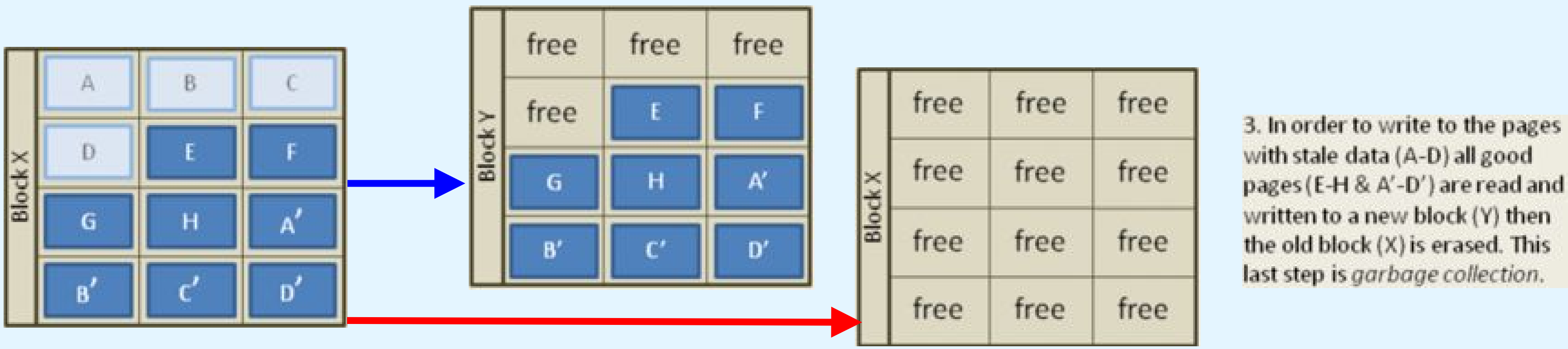
2. Four new pages (E-H) and four replacement pages (A′-D′) are written to the block (X). The original A-D pages are now invalid (stale) data, but cannot be overwritten until the whole block is erased.

Solution:

T = 4 x 250 us (write E,F,G,H) +
4 x 25 us (read A,B,C,D) +
4 x 250 us (write A',B',C',D') =
1000 us + 100 us + 1000 us = 2100 us = 2,1 ms

SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:

c)



Solution:

T = 8 x 25 us (read E, F, …, D’ from block X) +
8 x 250 us (write E, F, …, D’ in block Y) +
1 x 1500 us (erase block X) =
200 us + 2000 us + 1500 us = 3700 us = 3,7 ms